DESIGN AGGREGATE STRUCTURE

Supplemental Tool for ePAVE3 Program

OVERVIEW OF SUPERPAVE MIX DESIGN PROCESS

Step 1: Selection of Materials (Binder, Aggregate and Modifiers)

Step 2: Selection of DAS

Step 3: Selection of DAC

Step 4: Moisture Susceptibility Evaluation

Step 5: Rutting Resistance Evaluation

OVERVIEW OF SUPERPAVE MIX DESIGN PROCESS

Step 1: Selection of Materials (Binder, Aggregate and Modifiers)

Step 2: Selection of DAS → **Scope of the program**

Step 3: Selection of DAC

Step 4: Moisture Susceptibility Evaluation

Step 5: Rutting Resistance Evaluation

SAMPLE GRADATION

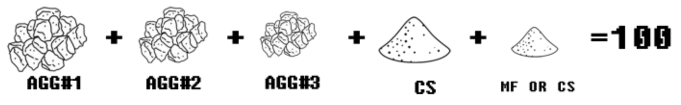
NMS: 25.0 mm

Step2: Selection of DAS

Once a group of aggregates has been identified, these aggregates are **combined at different percentages** to produce at least three distinct blends conforming to Superpave gradation requirements according to designed NMS.

4,598,126

POSSIBLE COMBINATION FOR 5 DIFFERENT STOCKPILE



View the Estimated VMA % of Each Blend

Input Values as Needed in the Green Cells

Back ...

Main Menu

25.0 mm Nominal Maximum Design

Next ...

Mix Design Reference No. 0

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Selection of DAS

Aggregate Gradation

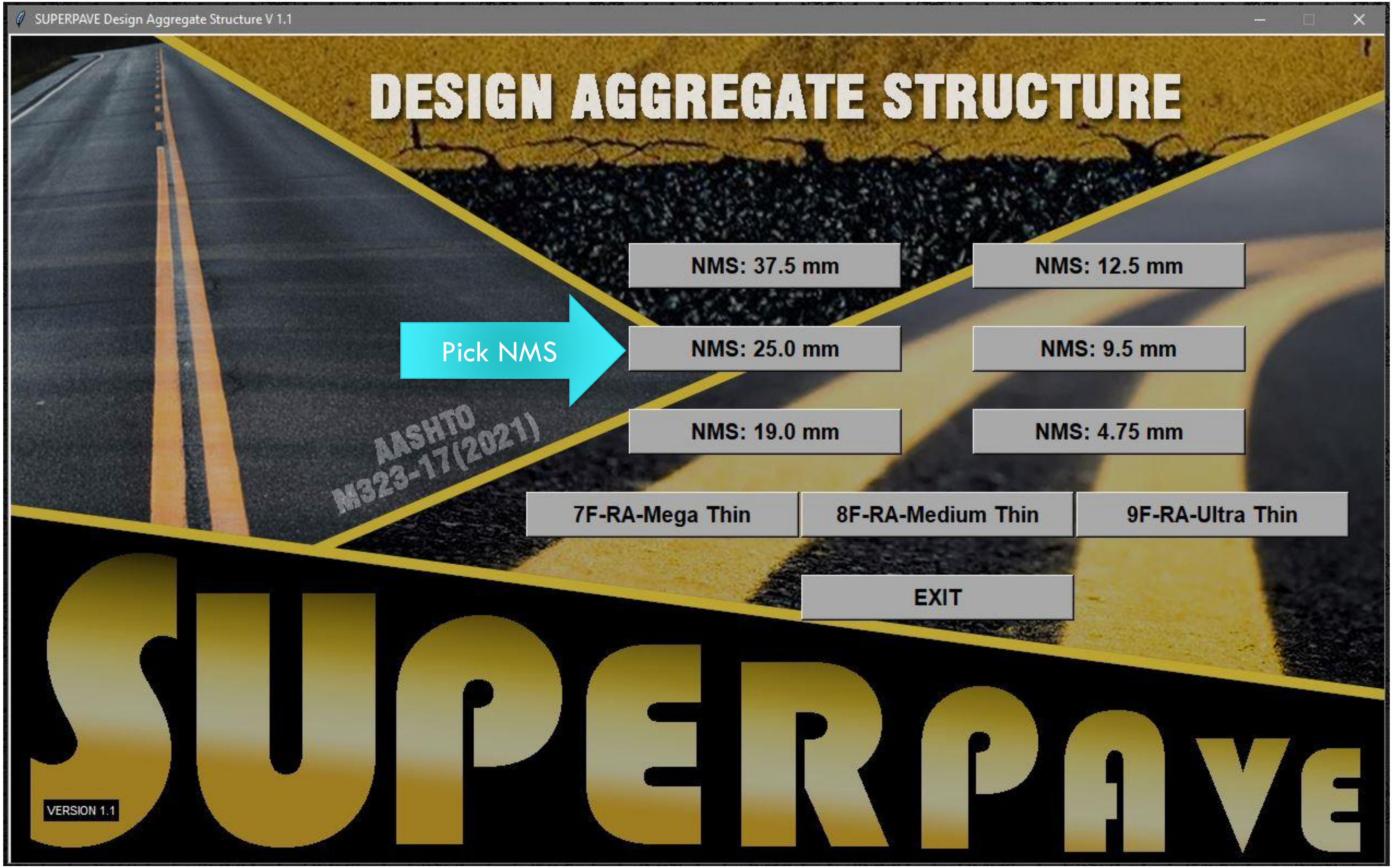
Sieve Size raised to 0.45 power	Sieve Size		Percentage Passing — Samples From the Hot Bins				
	mm	in	Agg. # 1	Agg. # 2	Agg. # 3	Agg. # 4	Agg. # 5
5.81	50.0	2	100.0	100.0	100.0	100.0	100.0
5.11	37.5	1.5	100.0	100.0	100.0	100.0	100.0
4.26	25	1	42.3	100.0	100.0	100.0	100.0
3.76	19.0	3/4	11.9	99.9	100.0	100.0	100.0
3.12	12.5	1/2	1.5	50.3	100.0	100.0	100.0
2.75	9.5	3/8	1.0	12.3	99.5	100.0	100.0
2.02	4.75	No.4	0.6	0.7	22.9	96.6	100.0
1.47	2.36	No 8	0.1	0.2	3.2	62.5	100.0
1.08	1.18	No 16	0.1	0.1	1.3	35.2	100.0
0.79	0.60	No 30	0.0	0.1	0.7	17.0	100.0
0.58	0.30	No 50	0.0	0.0	0.4	7.3	99.7
0.43	0.15	No 100	0.0	0.0	0.2	3.2	93.9
0.31	0.075	No. 200	0.0	0.0	0.1	1.1	77.2

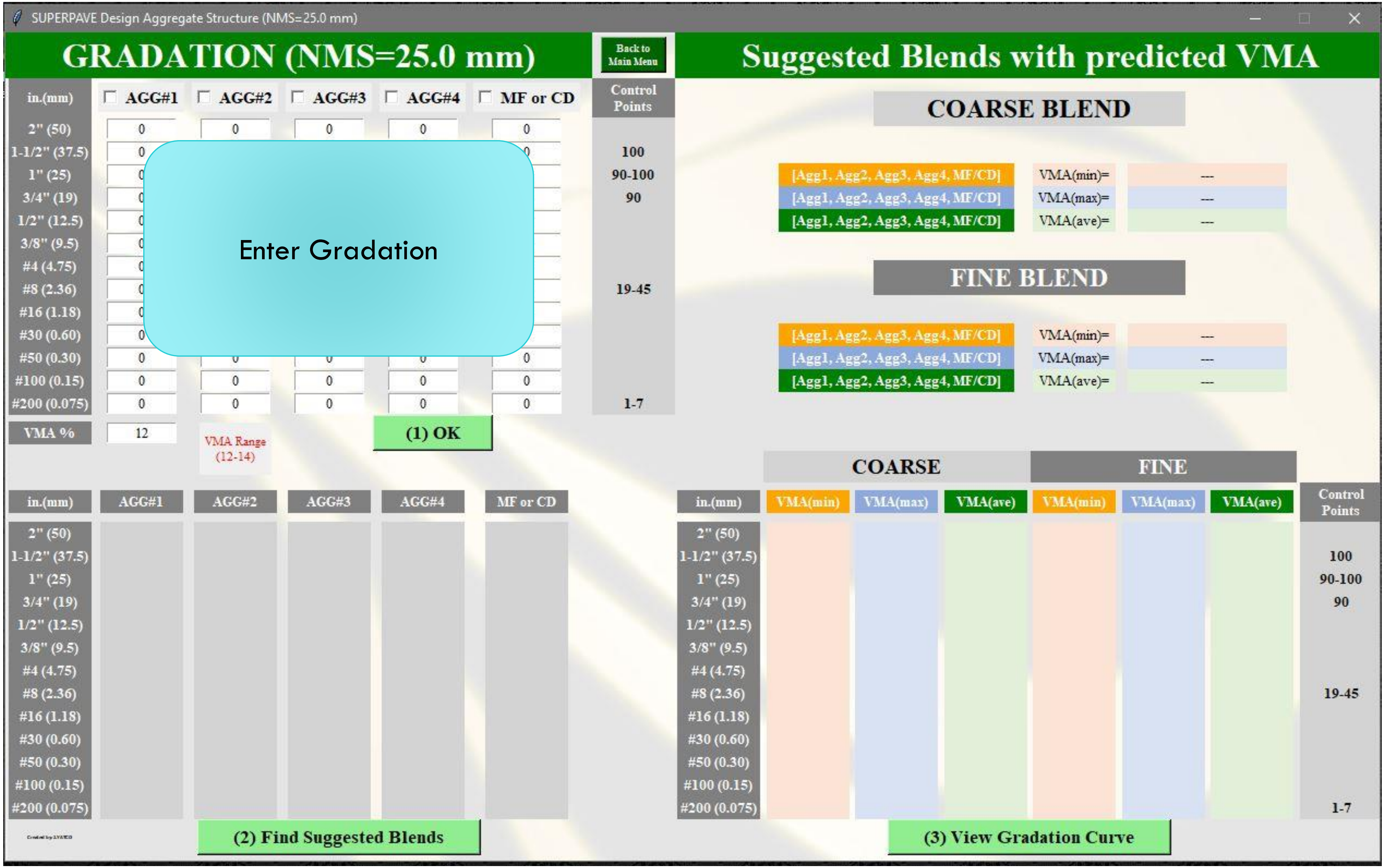
Suggested Blending Percentages

Blend #	Total	Agg. # 1	Agg. # 2	Agg. # 3	Agg. # 4	Agg. # 5
Blend 1	0.0					
Blend 2	0.0					
Blend 3	0.0					

USING THE PROGRAM

Demonstration of the program


SUPERPAVE Design Aggregate Structure V 1.1
DESIGN AGGREGATE STRUCTURE
Pick NMS
NMS: 37.5 mm
NMS: 12.5 mm
NMS: 25.0 mm
NMS: 9.5 mm
NMS: 19.0 mm
NMS: 4.75 mm
AASHTO
M323-17(2021)
7F-RA-Mega Thin
8F-RA-Medium Thin
9F-RA-Ultra Thin
EXIT
SUPERPAVE
VERSION 1.1


SUPERPAVE Design Aggregate Structure (NMS=25.0 mm)
GRADATION (NMS=25.0 mm)
Back to Main Menu
Suggested Blends with predicted VMA
in.(mm)
AGG#1
AGG#2
AGG#3
AGG#4
MF or CD
Control Points
2" (50)
1-1/2" (37.5)
1" (25)
3/4" (19)
1/2" (12.5)
3/8" (9.5)
#4 (4.75)
#8 (2.36)
#16 (1.18)
#30 (0.60)
#50 (0.30)
#100 (0.15)
#200 (0.075)
100
90-100
90
19-45
1-7
Enter Gradation
VMA %
12
VMA Range (12-14)
(1) OK
COARSE BLEND
[Agg1, Agg2, Agg3, Agg4, MF/CD]
VMA(min)=
VMA(max)=
VMA(ave)=
FINE BLEND
COARSE
FINE
VMA(min)
VMA(max)
VMA(ave)
(2) Find Suggested Blends
(3) View Gradation Curve

SUPERPAVE Design Aggregate Structure (NMS=25.0 mm)

GRADATION (NMS=25.0 mm)

Back to Main Menu

in.(mm)	AGG#1	AGG#2	AGG#3	AGG#4	MF or CD	Control Points
2" (50)	100	100	100	100	100	
1-1/2" (37.5)	100	100	100	100	100	100
1" (25)	42.3	100	100	100	100	90-100
3/4" (19)	11.9	99.9	100	100	100	90
1/2" (12.5)	1.5	50.3	100	100	100	
3/8" (9.5)	1	12.3	99.5	100	100	
#4 (4.75)	0.6	0.7	22.9	96.6	100	
#8 (2.36)	0.1	0.2	3.2	62.5	100	19-45
#16 (1.18)	0.1	0.1	1.3	35.2	100	
#30 (0.60)	0	0.1	0.7	17	100	
#50 (0.30)	0	0	0.4	7.3	99.7	
#100 (0.15)	0	0	0.2	3.2	93.9	
#200 (0.075)	0	0	0.1	1.1	77.2	1-7

VMA % 12

VMA Range (12-14)

(1) OK

Press OK

in.(mm)	AGG#1	AGG#2	AGG#3	AGG#4	MF or CD
2" (50)					
1-1/2" (37.5)					
1" (25)					
3/4" (19)					
1/2" (12.5)					
3/8" (9.5)					
#4 (4.75)					
#8 (2.36)					
#16 (1.18)					
#30 (0.60)					
#50 (0.30)					
#100 (0.15)					
#200 (0.075)					

(2) Find Suggested Blends

Suggested Blends with predicted VMA

COARSE BLEND

[Agg1, Agg2, Agg3, Agg4, MF/CD]	VMA(min)=	---
[Agg1, Agg2, Agg3, Agg4, MF/CD]	VMA(max)=	---
[Agg1, Agg2, Agg3, Agg4, MF/CD]	VMA(ave)=	---

FINE BLEND

[Agg1, Agg2, Agg3, Agg4, MF/CD]	VMA(min)=	---
[Agg1, Agg2, Agg3, Agg4, MF/CD]	VMA(max)=	---
[Agg1, Agg2, Agg3, Agg4, MF/CD]	VMA(ave)=	---

	COARSE			FINE			
in.(mm)	VMA(min)	VMA(max)	VMA(ave)	VMA(min)	VMA(max)	VMA(ave)	Control Points
2" (50)							
1-1/2" (37.5)							100
1" (25)							90-100
3/4" (19)							90
1/2" (12.5)							
3/8" (9.5)							
#4 (4.75)							
#8 (2.36)							19-45
#16 (1.18)							
#30 (0.60)							
#50 (0.30)							
#100 (0.15)							
#200 (0.075)							1-7

(3) View Gradation Curve

SUPERPAVE Design Aggregate Structure (NMS=25.0 mm)

GRADATION (NMS=25.0 mm)

Back to Main Menu

in.(mm)	AGG#1	AGG#2	AGG#3	AGG#4	MF or CD	Control Points
2" (50)	100	100	100	100	100	
1-1/2" (37.5)	100	100	100	100	100	100
1" (25)	42.3	100	100	100	100	90-100
3/4" (19)	11.9	99.9	100	100	100	90
1/2" (12.5)	1.5	50.3	100	100	100	
3/8" (9.5)	1	12.3	99.5	100	100	
#4 (4.75)	0.6	0.7	22.9	96.6	100	
#8 (2.36)	0.1	0.2	3.2	62.5	100	19-45
#16 (1.18)	0.1	0.1	1.3	35.2	100	
#30 (0.60)	0	0.1	0.7	17	100	
#50 (0.30)	0	0	0.4	7.3	99.7	
#100 (0.15)	0	0	0.2	3.2	93.9	
#200 (0.075)	0	0	0.1	1.1	77.2	1-7

VMA % 12

VMA Range (12-14)

(1) OK

in.(mm)	AGG#1	AGG#2	AGG#3	AGG#4	MF or CD
2" (50)	100.0	100.0	100.0	100.0	100.0
1-1/2" (37.5)	100.0	100.0	100.0	100.0	100.0
1" (25)	42.3	100.0	100.0	100.0	100.0
3/4" (19)	11.9	99.9	100.0	100.0	100.0
1/2" (12.5)	1.5	50.3	100.0	100.0	100.0
3/8" (9.5)	1.0	12.3	99.5	100.0	100.0
#4 (4.75)	0.6	0.7	22.9	96.6	100.0
#8 (2.36)	0.1	0.2	3.2	62.5	100.0
#16 (1.18)	0.1	0.1	1.3	35.2	100.0
#30 (0.60)	0.0	0.1	0.7	17.0	100.0
#50 (0.30)	0.0	0.0	0.4	7.3	99.7
#100 (0.15)	0.0	0.0	0.2	3.2	93.9
#200 (0.075)	0.00	0.00	0.10	1.10	77.20

(2) Find Suggested Blends

Check Data Entry then click Button(2)

Suggested Blends with predicted VMA

COARSE BLEND

[Agg1, Agg2, Agg3, Agg4, MF/CD]	VMA(min)=	---
[Agg1, Agg2, Agg3, Agg4, MF/CD]	VMA(max)=	---
[Agg1, Agg2, Agg3, Agg4, MF/CD]	VMA(ave)=	---

FINE BLEND

[Agg1, Agg2, Agg3, Agg4, MF/CD]	VMA(min)=	---
[Agg1, Agg2, Agg3, Agg4, MF/CD]	VMA(max)=	---
[Agg1, Agg2, Agg3, Agg4, MF/CD]	VMA(ave)=	---

	COARSE			FINE			
in.(mm)	VMA(min)	VMA(max)	VMA(ave)	VMA(min)	VMA(max)	VMA(ave)	Control Points
2" (50)							
1-1/2" (37.5)							100
1" (25)							90-100
3/4" (19)							90
1/2" (12.5)							
3/8" (9.5)							
#4 (4.75)							
#8 (2.36)							19-45
#16 (1.18)							
#30 (0.60)							
#50 (0.30)							
#100 (0.15)							
#200 (0.075)							1-7

(3) View Gradation Curve

SUPERPAVE Design Aggregate Structure (NMS=25.0 mm)

GRADATION (NMS=25.0 mm)

Back to Main Menu

in.(mm)	AGG#1	AGG#2	AGG#3	AGG#4	MF or CD	Control Points
2" (50)	100	100	100	100	100	
1-1/2" (37.5)	100	100	100	100	100	100
1" (25)	42.3	100	100	100	100	90-100
3/4" (19)	11.9	99.9	100	100	100	90
1/2" (12.5)	1.5	50.3	100	100	100	
3/8" (9.5)	1	12.3	99.5	100	100	
#4 (4.75)	0.6	0.7	22.9	96.6	100	
#8 (2.36)	0.1	0.2	3.2	62.5	100	19-45
#16 (1.18)	0.1	0.1	1.3	35.2	100	
#30 (0.60)	0	0.1	0.7	17	100	
#50 (0.30)	0	0	0.4	7.3	99.7	
#100 (0.15)	0	0	0.2	3.2	93.9	
#200 (0.075)	0	0	0.1	1.1	77.2	1-7

VMA % 12

VMA Range (12-14)

(1) OK

Suggested Blends with predicted VMA

COARSE BLEND

14 30 20 30 6	VMA(min)=	12.71
14 15 40 25 6	VMA(max)=	13.97
16 25 30 25 4	VMA(ave)=	13.34

FINE BLEND

14 25 25 30 6	VMA(min)=	12.82
12 25 25 35 3	VMA(max)=	13.76
16 20 30 30 4	VMA(ave)=	13.26

in.(mm)	AGG#1	AGG#2	AGG#3	AGG#4	MF or CD
2" (50)	100.0	100.0	100.0	100.0	100.0
1-1/2" (37.5)	100.0	100.0	100.0	100.0	100.0
1" (25)	42.3	100.0	100.0	100.0	100.0
3/4" (19)	11.9	99.9	100.0	100.0	100.0
1/2" (12.5)	1.5	50.3	100.0	100.0	100.0
3/8" (9.5)	1.0	12.3	99.5	100.0	100.0
#4 (4.75)	0.6	0.7	22.9	96.6	100.0
#8 (2.36)	0.1	0.2	3.2	62.5	100.0
#16 (1.18)	0.1	0.1	1.3	35.2	100.0
#30 (0.60)	0.0	0.1	0.7	17.0	100.0
#50 (0.30)	0.0	0.0	0.4	7.3	99.7
#100 (0.15)	0.0	0.0	0.2	3.2	93.9
#200 (0.075)	0.00	0.00	0.10	1.10	77.20

(2) Find Suggested Blends

	COARSE			FINE			
in.(mm)	VMA(min)	VMA(max)	VMA(ave)	VMA(min)	VMA(max)	VMA(ave)	Control Points
2" (50)	100.0	100.0	100.0	100.0	100.0	100.0	
1-1/2" (37.5)	100.0	100.0	100.0	100.0	100.0	100.0	100
1" (25)	91.9	91.9	90.8	91.9	93.1	90.8	90-100
3/4" (19)	87.6	87.7	85.9	87.6	89.4	85.9	90
1/2" (12.5)	71.3	78.8	71.8	73.8	75.8	74.3	
3/8" (9.5)	59.7	72.8	62.1	64.1	66.1	66.5	
#4 (4.75)	39.9	39.5	35.3	41.0	42.8	40.1	
#8 (2.36)	25.5	22.9	20.7	25.6	25.7	23.8	19-45
#16 (1.18)	16.9	15.3	13.2	16.9	15.7	15.0	
#30 (0.60)	11.3	10.5	8.5	11.3	9.2	9.3	
#50 (0.30)	8.3	8.0	5.9	8.3	5.6	6.3	
#100 (0.15)	6.6	6.5	4.6	6.6	4.0	4.8	
#200 (0.075)	4.98	4.95	3.39	4.99	2.73	3.45	1-7

(3) View Gradation Curve

SUPERPAVE Design Aggregate Structure (NMS=25.0 mm)

GRADATION (NMS=25.0 mm)

Back to Main Menu

in.(mm)	AGG#1	AGG#2	AGG#3	AGG#4	MF or CD	Control Points
2" (50)	100	100	100	100	100	
1-1/2" (37.5)	100	100	100	100	100	100
1" (25)	42.3	100	100	100	100	90-100
3/4" (19)	11.9	99.9	100	100	100	90
1/2" (12.5)	1.5	50.3	100	100	100	
3/8" (9.5)	1	12.3	99.5	100	100	
#4 (4.75)	0.6	0.7	22.9	96.6	100	
#8 (2.36)	0.1	0.2	3.2	62.5	100	19-45
#16 (1.18)	0.1	0.1	1.3	35.2	100	
#30 (0.60)	0	0.1	0.7	17	100	
#50 (0.30)	0	0	0.4	7.3	99.7	
#100 (0.15)	0	0	0.2	3.2	93.9	
#200 (0.075)	0	0	0.1	1.1	77.2	1-7

VMA % 12

VMA Range (12-14)

(1) OK

Suggested Blends with predicted VMA

COARSE BLEND

14 30 20 30 6	VMA(min)=	12.71
14 15 40 25 6	VMA(max)=	13.97
16 25 30 25 4	VMA(ave)=	13.34

FINE BLEND

14 25 25 30 6	VMA(min)=	12.82
12 25 25 35 3	VMA(max)=	13.76
16 20 30 30 4	VMA(ave)=	13.26

in.(mm)	AGG#1	AGG#2	AGG#3	AGG#4	MF or CD
2" (50)	100.0	100.0	100.0	100.0	100.0
1-1/2" (37.5)	100.0	100.0	100.0	100.0	100.0
1" (25)	42.3	100.0	100.0	100.0	100.0
3/4" (19)	11.9	99.9	100.0	100.0	100.0
1/2" (12.5)	1.5	50.3	100.0	100.0	100.0
3/8" (9.5)	1.0	12.3	99.5	100.0	100.0
#4 (4.75)	0.6	0.7	22.9	96.6	100.0
#8 (2.36)	0.1	0.2	3.2	62.5	100.0
#16 (1.18)	0.1	0.1	1.3	35.2	100.0
#30 (0.60)	0.0	0.1	0.7	17.0	100.0
#50 (0.30)	0.0	0.0	0.4	7.3	99.7
#100 (0.15)	0.0	0.0	0.2	3.2	93.9
#200 (0.075)	0.00	0.00	0.10	1.10	77.20

(2) Find Suggested Blends

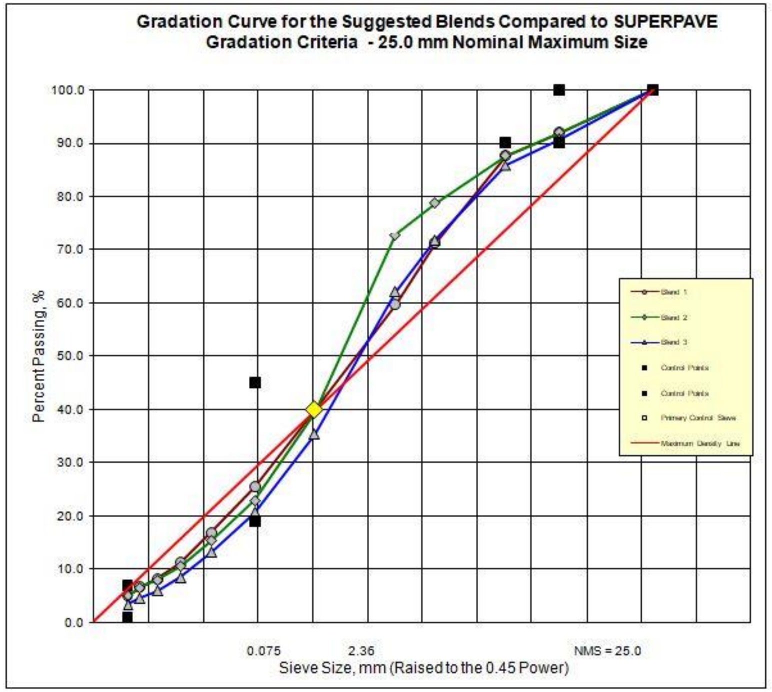
	COARSE			FINE			
in.(mm)	VMA(min)	VMA(max)	VMA(ave)	VMA(min)	VMA(max)	VMA(ave)	Control Points
2" (50)	100.0	100.0	100.0	100.0	100.0	100.0	
1-1/2" (37.5)	100.0	100.0	100.0	100.0	100.0	100.0	100
1" (25)	91.9	91.9	90.8	91.9	93.1	90.8	90-100
3/4" (19)	87.6	87.7	85.9	87.6	89.4	85.9	90
1/2" (12.5)	71.3	78.8	71.8	73.8	75.8	74.3	
3/8" (9.5)	59.7	72.8	62.1	64.1	66.1	66.5	
#4 (4.75)	39.9	39.5	35.3	41.0	42.8	40.1	
#8 (2.36)	25.5	22.9	20.7	25.6	25.7	23.8	19-45
#16 (1.18)	16.9	15.3	13.2	16.9	15.7	15.0	
#30 (0.60)	11.3	10.5	8.5	11.3	9.2	9.3	
#50 (0.30)	8.3	8.0	5.9	8.3	5.6	6.3	
#100 (0.15)	6.6	6.5	4.6	6.6	4.0	4.8	
#200 (0.075)	4.98	4.95	3.39	4.99	2.73	3.45	1-7

(3) View Gradation Curve

SUGGESTED COARSE BLEND

COARSE BLEND

14	30	20	30	6	VMA(min)=	12.71
14	15	40	25	6	VMA(max)=	13.97
16	25	30	25	4	VMA(ave)=	13.34



Suggested Blending Percentages

Blend #	Total	Agg. # 1	Agg. # 2	Agg. # 3	Agg. # 4	Agg. # 5
Blend 1	100.0	14.0	30.0	20.0	30.0	6.0
Blend 2	100.0	14.0	15.0	40.0	25.0	6.0
Blend 3	100.0	16.0	25.0	30.0	25.0	4.0

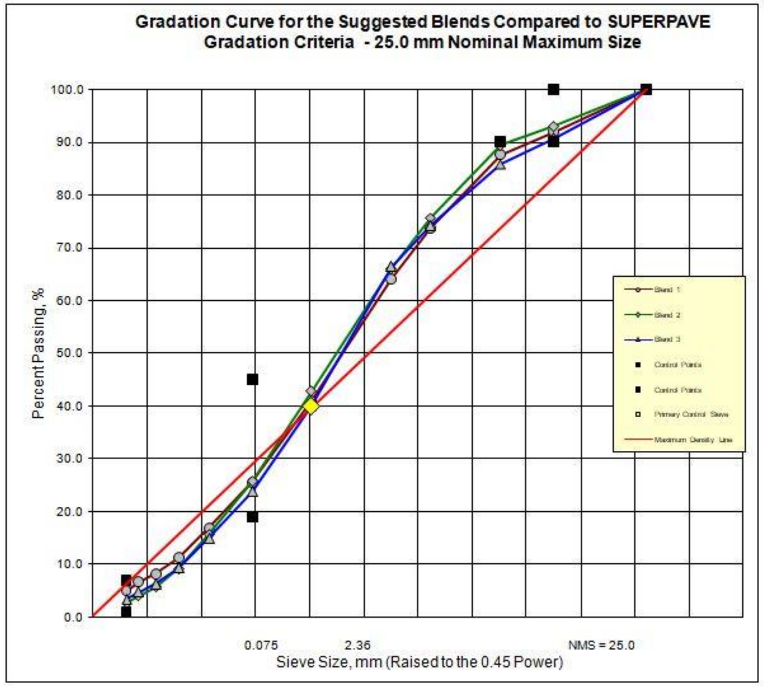
Selection of DAS

Sieve Size raised to 0.45 power	Sieve Size in mm	SUPERPAVE Criteria, % Passing			% Passing			
		Control Points		The Primary Control Sieve				
		Min	Max		Blend 1	Blend 2	Blend 3	Remarks
5.81	50.0			The PCS was introduced in the new AASHTO M323 "Superpave Volumetric Mix Design Specifications", 2005. It is defined as the sieve size that classifies the gradation between course and fine. PCS is specified for each NMS. For 25.0 mm NMS the PCS is 4.75 mm and the % passing at this sieve is 40.0%. If the gradation line passes above the PCS then the gradation is classified as fine otherwise it is classified as coarse.	100.0	100.0	100.0	
5.11	37.5	100.0			100.0	100.0	100.0	
4.26	25.0	90.0	100.0		91.9	91.9	90.8	
3.76	19.0		90.0		87.6	87.7	85.9	
3.12	12.5				71.3	78.8	71.8	
2.75	9.5				59.7	72.8	62.1	
2.02	4.75				39.9	39.5	35.3	
1.47	2.36	19.0	45.0		25.5	22.9	20.7	
1.08	1.18				16.9	15.3	13.2	
0.79	0.60				11.3	10.5	8.5	
0.58	0.30				8.3	8.0	5.9	
0.43	0.15				6.6	6.5	4.6	
0.31	0.075	1.0	7.0		5.0	4.9	3.4	
Based on the PCS Criteria, the gradation of each blend is classified as					Coarse	Coarse	Coarse	

SUGGESTED FINE BLEND

FINE BLEND

14 25 25 30 6	VMA(min)=	12.82
12 25 25 35 3	VMA(max)=	13.76
16 20 30 30 4	VMA(ave)=	13.26



Suggested Blending Percentages

Blend #	Total	Agg. # 1	Agg. # 2	Agg. # 3	Agg. # 4	Agg. # 5
Blend 1	100.0	14.0	25.0	25.0	30.0	6.0
Blend 2	100.0	12.0	25.0	25.0	35.0	3.0
Blend 3	100.0	16.0	20.0	30.0	30.0	4.0

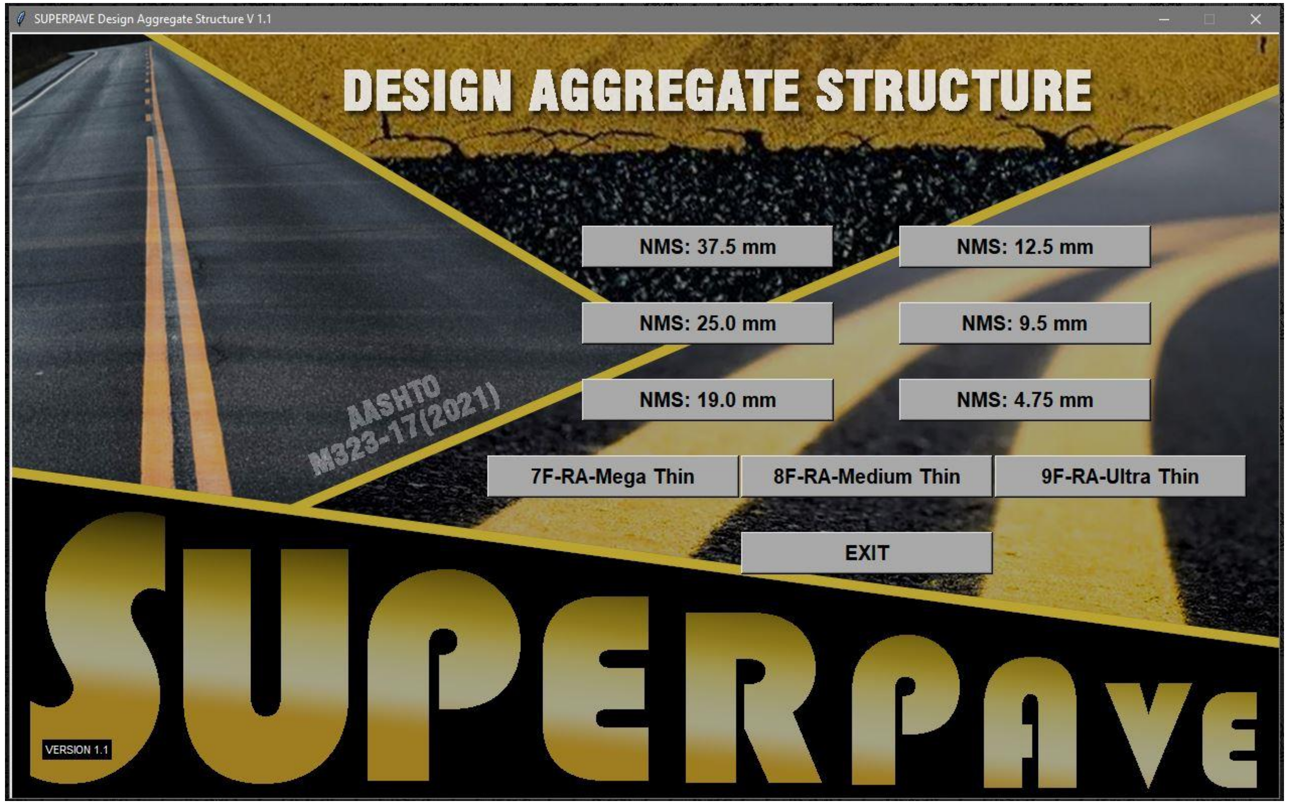
Selection of DAS

Sieve Size raised to 0.45 power	Sieve Size in mm	SUPERPAVE Criteria, % Passing			% Passing			
		Control Points		The Primary Control Sieve				
		Min	Max		Blend 1	Blend 2	Blend 3	Remarks
5.81	50.0			The PCS was introduced in the new AASHTO M323 "Superpave Volumetric Mix Design Specifications", 2005. It is defined as the sieve size that classifies the gradation between course and fine. PCS is specified for each NMS. For 25.0 mm NMS the PCS is 4.75 mm and the % passing at this sieve is 40.0%. If the gradation line passes above the PCS then the gradation is classified as fine otherwise it is classified as coarse.	100.0	100.0	100.0	
5.11	37.5	100.0			100.0	100.0	100.0	
4.26	25.0	90.0	100.0		91.9	93.1	90.8	
3.76	19.0		90.0		87.6	89.4	85.9	
3.12	12.5				73.8	75.8	74.3	
2.75	9.5				64.1	66.1	66.5	
2.02	4.75				41.0	42.8	40.1	
1.47	2.36	19.0	45.0		25.6	25.7	23.8	
1.08	1.18				16.9	15.7	15.0	
0.79	0.60				11.3	9.2	9.3	
0.58	0.30				8.3	5.6	6.3	
0.43	0.15				6.6	4.0	4.8	
0.31	0.075	1.0	7.0		5.0	2.7	3.4	
Based on the PCS Criteria, the gradation of each blend is classified as					Fine	Fine	Fine	

PURPOSE

This program suggests initial blend of aggregate that conforms to control points or gradation requirements by AASHTO M323 (Superpave).

This program eliminates trial-and-error of different combination and provide distinct blends for the design



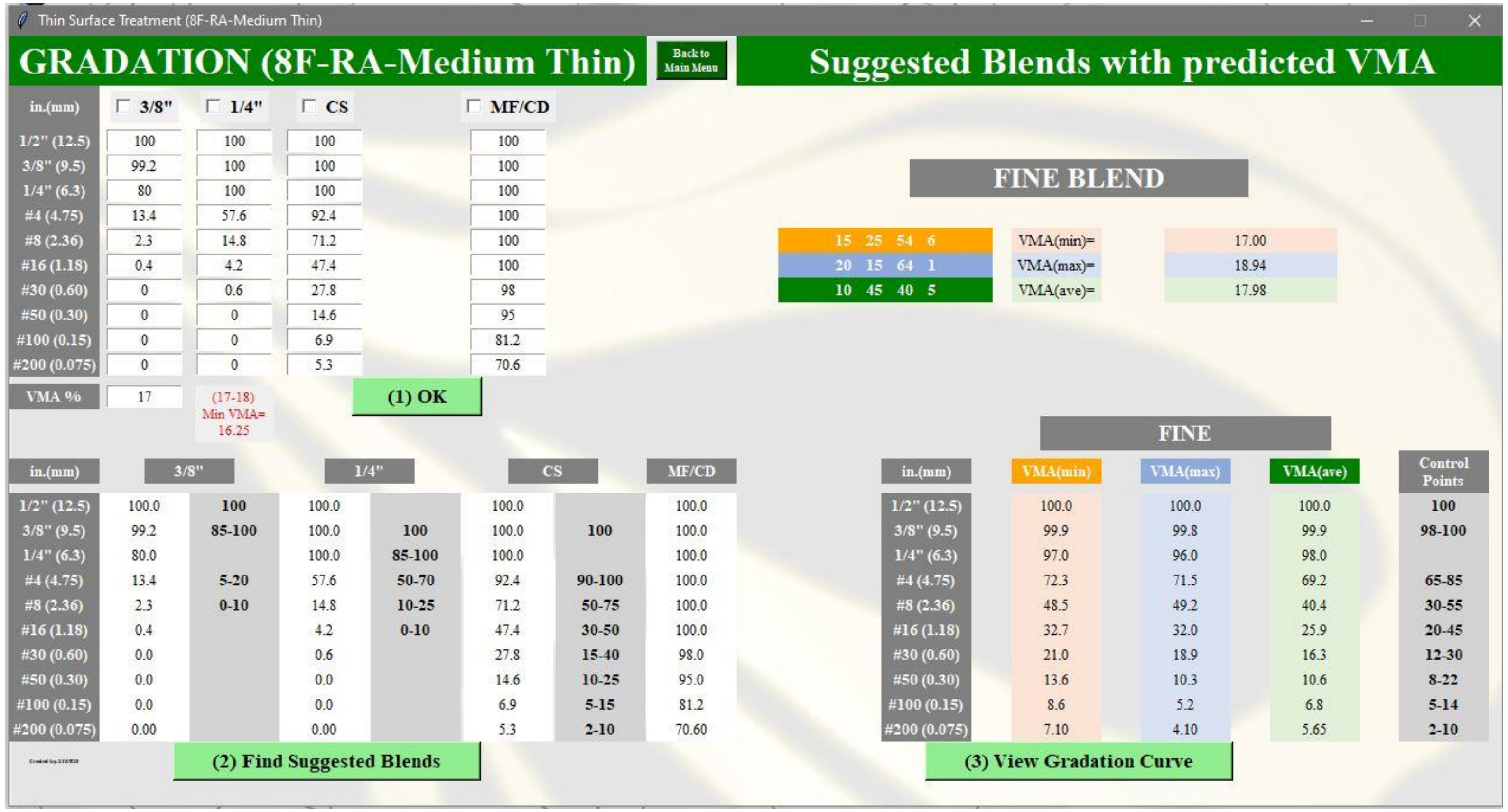
BONUS

Thin Layer Design

BONUS

<u>MOT TECHNICAL MANUAL:</u>

FOR THE DEVELOPED ASPHALT MIXTURES USED WITHIN THIN LAYERS OF SURFACE TREATMENT FOR ROAD MAINTENANCE



in.(mm)	3/8"	1/4"	CS	MF/CD
1/2" (12.5)	100	100	100	100
3/8" (9.5)	99.2	100	100	100
1/4" (6.3)	80	100	100	100
#4 (4.75)	13.4	57.6	92.4	100
#8 (2.36)	2.3	14.8	71.2	100
#16 (1.18)	0.4	4.2	47.4	100
#30 (0.60)	0	0.6	27.8	98
#50 (0.30)	0	0	14.6	95
#100 (0.15)	0	0	6.9	81.2
#200 (0.075)	0	0	5.3	70.6
VMA %	17	(17-18) Min VMA= 16.25		

(1) OK

FINE BLEND

15 25 54 6	VMA(min)=	17.00
20 15 64 1	VMA(max)=	18.94
10 45 40 5	VMA(ave)=	17.98

in.(mm)	3/8"		1/4"		CS		MF/CD
1/2" (12.5)	100.0	100	100.0		100.0		100.0
3/8" (9.5)	99.2	85-100	100.0	100	100.0	100	100.0
1/4" (6.3)	80.0		100.0	85-100	100.0		100.0
#4 (4.75)	13.4	5-20	57.6	50-70	92.4	90-100	100.0
#8 (2.36)	2.3	0-10	14.8	10-25	71.2	50-75	100.0
#16 (1.18)	0.4		4.2	0-10	47.4	30-50	100.0
#30 (0.60)	0.0		0.6		27.8	15-40	98.0
#50 (0.30)	0.0		0.0		14.6	10-25	95.0
#100 (0.15)	0.0		0.0		6.9	5-15	81.2
#200 (0.075)	0.00		0.00		5.3	2-10	70.60

(2) Find Suggested Blends

FINE

in.(mm)	VMA(min)	VMA(max)	VMA(ave)	Control Points
1/2" (12.5)	100.0	100.0	100.0	100
3/8" (9.5)	99.9	99.8	99.9	98-100
1/4" (6.3)	97.0	96.0	98.0	
#4 (4.75)	72.3	71.5	69.2	65-85
#8 (2.36)	48.5	49.2	40.4	30-55
#16 (1.18)	32.7	32.0	25.9	20-45
#30 (0.60)	21.0	18.9	16.3	12-30
#50 (0.30)	13.6	10.3	10.6	8-22
#100 (0.15)	8.6	5.2	6.8	5-14
#200 (0.075)	7.10	4.10	5.65	2-10

(3) View Gradation Curve